

Are multi-modal methods as effective as constraint in aphasia treatment?

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Background

Constraint

Constraint Induced Aphasia Therapy (CIAT) is based on the success of Constraint Induced Movement Therapy. Constraint approaches emphasise **speech cues and production** in therapy because nonverbal actions such as gesture or drawing are thought to contribute to **learned nonuse**.



Multimodal

Multimodal aphasia treatments capitalise on the rich multimodal nature of human communication and use modalities such as gesture, drawing, writing and music to **facilitate** word finding.



Despite being theoretically opposed, **both approaches have demonstrated success** in treating post-stroke aphasia. A comparison of the two was sought as it has important implications both clinically and theoretically.

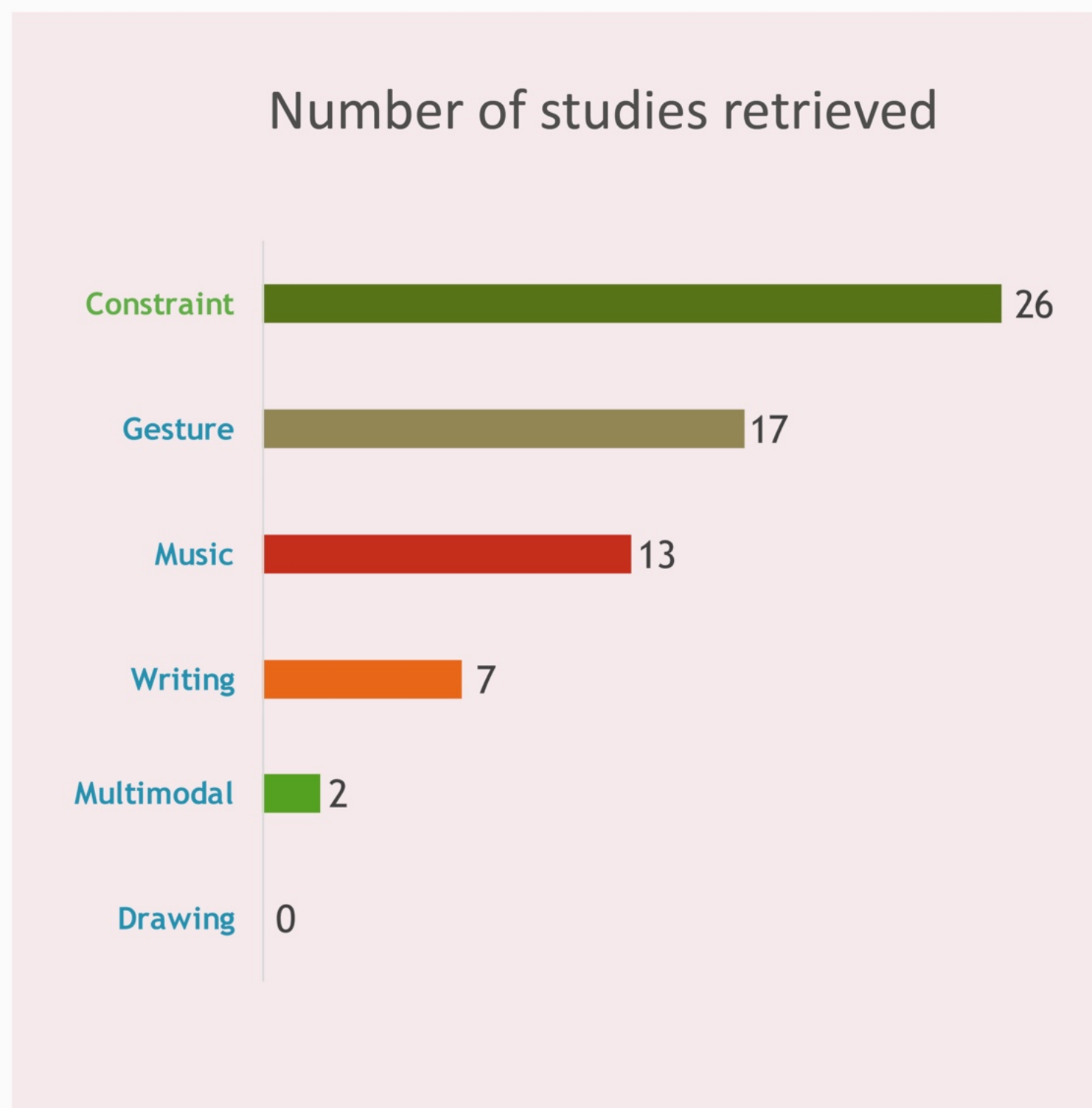
Aims

To compare the effects of **constraint** and **multimodal** therapies in people with chronic aphasia at the levels of impairment, participation and quality of life.

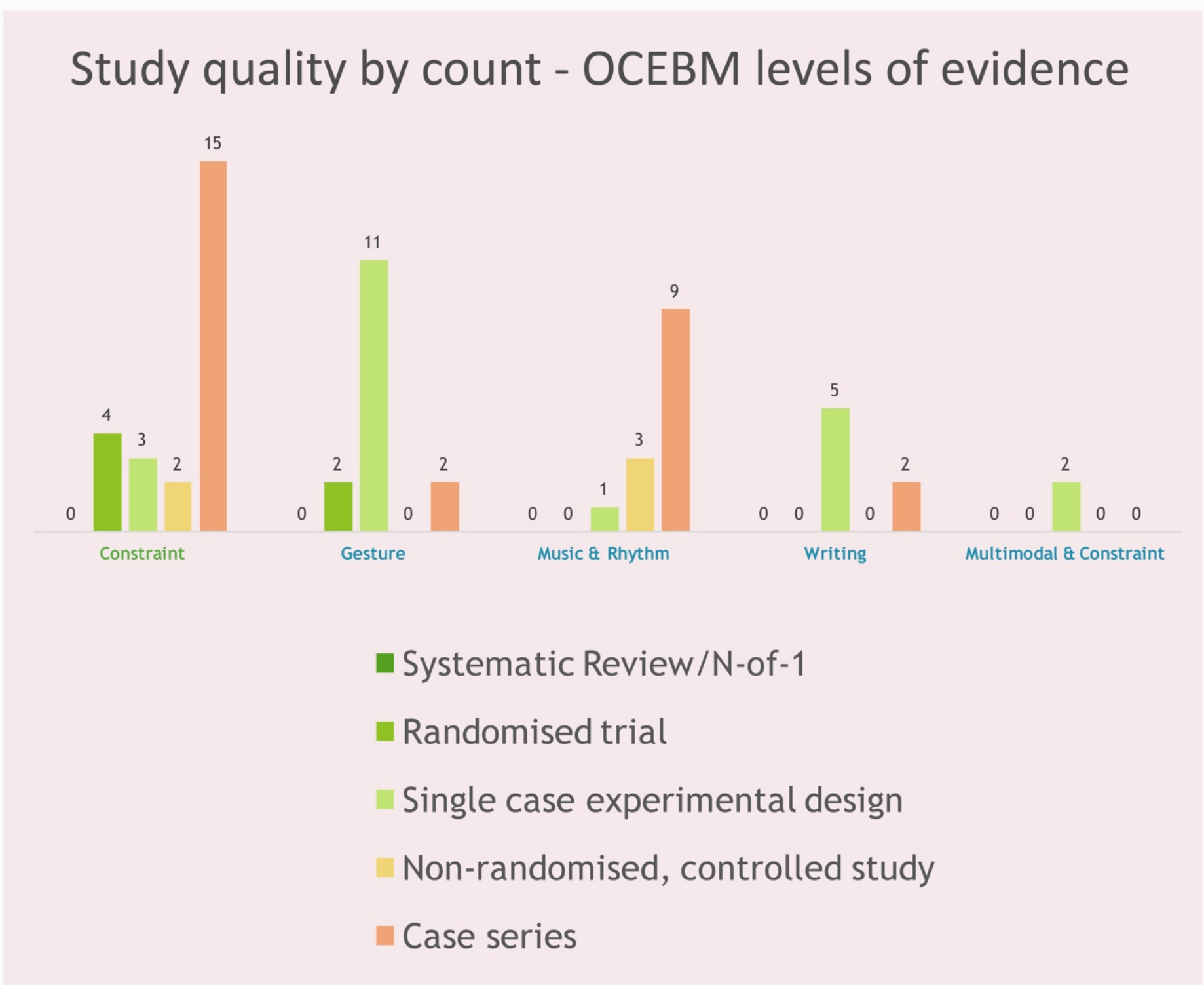
Methods

A systematic search of major databases was conducted. Two authors screened results and rated the level of evidence and methodological quality (using PEDro-P and RoBINT).

Results



Fifty-nine papers were identified that met criteria – 24 used **constraint** and 37 used **multimodal** treatment, while two compared **constraint** and **multimodal** therapies directly. Meta-analysis was not possible due to heterogeneity. Results were broadly positive for both treatments with neither being clearly superior; however, high quality research was limited. Most studies looked at impairment level outcomes without considering participation or quality of life. The application of **constraint** also varied significantly between studies.

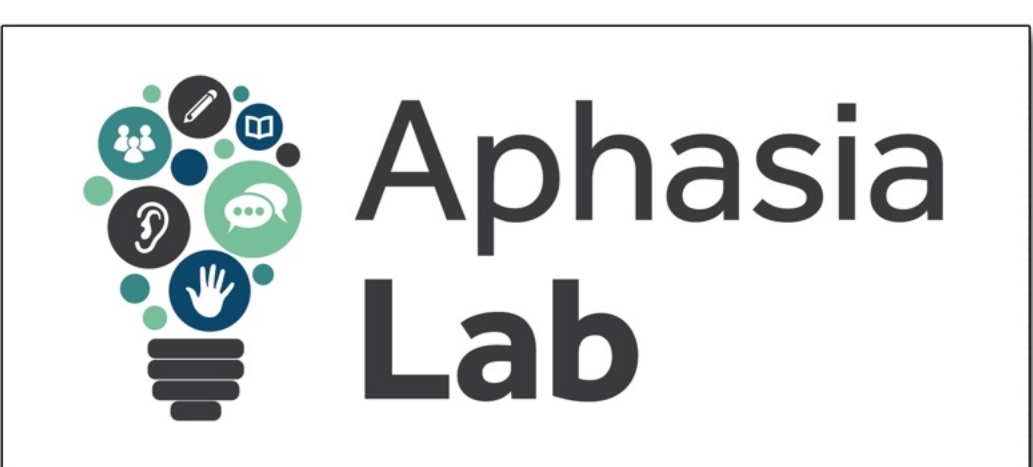


Conclusion

- There is insufficient evidence to demonstrate superiority of **constraint** or **multimodal** treatments
- There is limited high quality evidence for **constraint** and **multimodal** therapies for chronic aphasia
- Some **multimodal** treatments such as drawing and writing have limited evidence of any quality
- Most high level studies on **constraint** confound with intensity or do not compare to **non-constraint** treatments
- Few studies investigated quality of life or participation in life activities
- The interpretation of '**constraint**' requires clearer reporting in future

Future research required:

1. Direct, rigorous comparison of **constraint** and **multimodal** aphasia therapies
2. Further research on **quality of life** and **participation** outcomes
3. More rigorous data on **constraint** and **multimodal** therapies vs controls



Interested?
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